CALORIE METER-FRUITS AND VEGETABLES CALORIE ESTIMATION USING MACHINE LEARNING

ABSTRACT

Machine Learning (ML) is a very powerful and important technology in the world today. With the help of ML modules, various appropriate algorithms such as Faster RCNN algorithm, canny edge detection algorithm and GrabCut segmentation algorithm are applied to the proposed system. This system focuses mainly on the calculation of calories and other nutrients present in food. The whole thing will be automated as opposed to existing systems where the user needs to manually deliver the values. However, users will only need to click on the food image and provide it as an input to the system. Further processes can be automated quickly, such as the use of nearer R-CNN to perceive for each food and standardization item. The GrabCut algorithm is used to get the outline of each food. Then the volume of individually food is determined by formulas for volume valuation. Lastly, it estimates the calories of each food and experimental studies have shown that by providing production with information of calories and nutrients present in the food, the proposed estimation method is effective.

1. INTRODUCTION

1.1 Overview

In the present digital age, where technology has made rapid advancements in almost every domain, health and nutrition remain critical global priorities. Despite the availability of medical knowledge and nutritional guidelines, many individuals struggle to maintain a balanced diet. Modern lifestyles, characterized by busy work schedules, urban living, and fast food consumption, have led to poor dietary habits. This shift has contributed to a rise in obesity, malnutrition, and chronic diseases such as diabetes, cardiovascular disorders, and certain forms of cancer. The World Health Organization (WHO) reports that unhealthy diets are a leading factor in global mortality rates. In 2016 alone, statistical data revealed that 39% of adults aged 18 years and above were overweight, while 13% were classified as obese. Alarmingly, more people today die from overweight and obesity-related conditions than from being underweight.

One of the major challenges in promoting healthier lifestyles is the lack of accessible, accurate, and convenient tools for tracking food consumption. Traditional calorie counting

methods often rely on manual data entry — requiring the user to input the type of food, portion size, or weight. While effective in theory, these methods are tedious, prone to human error, and often lead to inconsistent or incomplete records. Furthermore, individuals may not always be aware of the nutritional composition of certain foods, especially in cases where portion sizes are difficult to estimate visually. This creates a gap between awareness and actionable health management.

The Calorie Meter – Fruits and Vegetables Calorie Estimation Using Machine Learning project aims to address this gap by providing a fully automated solution that leverages modern technological advancements. Instead of requiring manual entry, the system allows users to simply upload images of their food items — specifically fruits and vegetables — to instantly receive accurate calorie estimates. The approach integrates deep learning models, specifically a Convolutional Neural Network (CNN), to classify the food item based on its visual characteristics. For improved accuracy, the system recommends capturing two views of the item: the top view and the front view.

Once the food item is recognized, the system connects to the Nutritionix API, a trusted and extensive nutritional database, to fetch real-time calorie and nutrient information. This not only provides instant results but also ensures that the nutritional data remains updated with the latest values. The results are then displayed to the user, along with the image they uploaded. All predictions are stored in the user's profile, allowing them to review their past dietary records and monitor calorie intake over time.

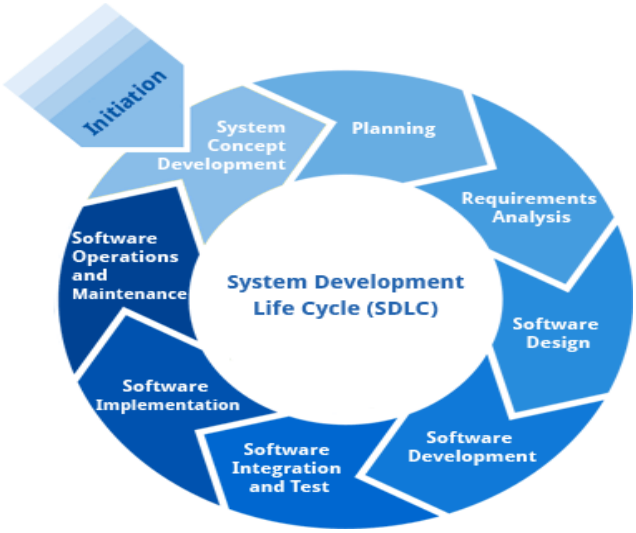
The system is implemented as a web application using the Django's framework, offering a user-friendly interface and secure login features. It supports functionalities such as user registration, profile management, image upload, prediction display, and history tracking. The machine learning component is trained on a carefully curated dataset containing 36 classes of fruits and vegetables, ensuring broad coverage and high classification accuracy.

The significance of this project lies not only in its technical innovation but also in its potential societal impact. By simplifying the process of calorie estimation, it encourages users to be more conscious of their food intake and make healthier choices. It also opens opportunities for integration with other health monitoring systems, fitness applications, and dietary planning tools. Ultimately, the Calorie Meter project is a step toward bridging the gap between technological capability and practical health management, providing a solution that is accessible, accurate, and efficient for everyday use.

2. LITERATURE OVERVIEW

2.1 The Systems Development Life Cycle (SDLC):

The Systems Development Life Cycle (SDLC), or Software Development Life Cycle in systems engineering, information systems and software engineering, is the process of creating or altering systems, and the models and methodologies use to develop these systems.



2.1.1 Requirement Analysis and Design:

Analysis gathers the requirements for the system. This stage includes a detailed study of the business needs of the organization. Options for changing the business process may be considered. Design focuses on high level design like, what programs are needed and how are they going to interact, low-level design (how the individual programs are going to work), interface design (what are the interfaces going to look like) and data design (what data will be required). During these phases, the software's overall structure is defined. Analysis and Design are very crucial in the whole development cycle. Any glitch in the design phase could be very expensive to solve in the later stage of the software development. Much care is taken during this phase. The logical system of the product is developed in this phase.

2.1.2 Implementation:

In this phase the designs are translated into code. Computer programs are written using a conventional programming language or an application generator. Programming tools like Compilers, Interpreters, and Debuggers are used to generate the code. Different high level programming languages like PYTHON 3.6, Anaconda Cloud are used for coding. With respect to the type of application, the right programming language is chosen.

2.1.3 Testing:

In this phase the system is tested. Normally programs are written as a series of individual modules, this subject to separate and detailed test. The system is then tested as a whole. The separate modules are brought together and tested as a complete system. The system is tested to ensure that interfaces between modules work (integration testing), the system works on the intended platform and with the expected volume of data (volume testing) and that the system does what the user requires (acceptance/beta testing).

2.1.4 Maintenance:

Inevitably the system will need maintenance. Software will definitely undergo change once it is delivered to the customer. There are many reasons for the change. Change could happen because of some unexpected input values into the system. In addition, the changes in the system could directly affect the software operations. The software should be developed to accommodate changes that could happen during the post implementation period.

2.2 Software Developing Process Model Waterfall Model:

The Waterfall Model was the first Process Model to be introduced. It is also referred to as a linear-sequential life cycle model. It is very simple to understand and use. In a waterfall model, each phase must be completed before the next phase can begin and there is no overlapping in the phases.

The Waterfall model is the earliest SDLC approach that was used for software development. The waterfall Model illustrates the software development process in a linear sequential flow. This means that any phase in the development process begins only if the previous phase is complete. In this waterfall model, the phases do not overlap. Waterfall

Model – Design Waterfall approach was first SDLC Model to be used widely in Software Engineering to ensure success of the project. In "The Waterfall" approach, the whole process of software development is divided into separate phases. In this Waterfall model, typically, the outcome of one phase acts as the input for the next phase sequentially. The following illustration is a representation of the different phases of the Waterfall Model.

Waterfall Model – Design:

Waterfall approach was first SDLC Model to be used widely in Software Engineering to ensure success of the project. In "The Waterfall" approach, the whole process of software development is divided into separate phases. In this Waterfall model, typically, the outcome of one phase acts as the input for the next phase sequentially. The following illustration is a representation of the different phases of the Waterfall Model.

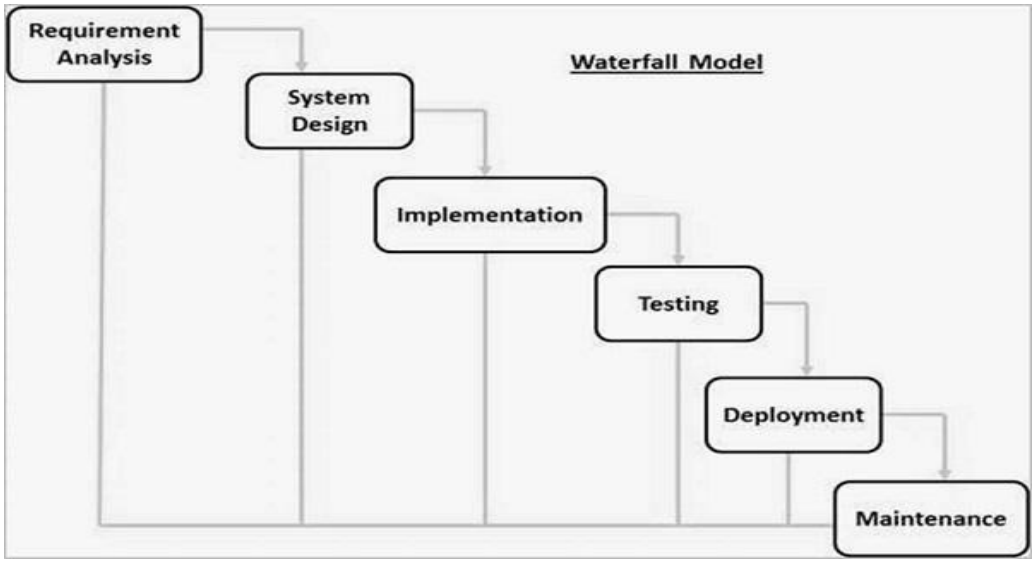


Figure1.Different phases of waterfall model

The sequential phases in Waterfall model are

- **Requirement Gathering and analysis** − All possible requirements of the system to be developed are captured in this phase and documented in a requirement specification document.

- **System Design** − the requirement specifications from first phase are studied in this phase and the system design is prepared. This system design helps in specifying hardware and system requirements and helps in defining the overall system architecture.

- **Implementation** − with inputs from the system design, the system is first developed in small programs called units, which are integrated in the next phase. Each unit is developed and tested for its functionality, which is referred to as Unit Testing.

- **Integration and Testing** − All the units developed in the implementation phase are integrated into a system after testing of each unit. Post integration the entire system is tested for any faults and failures.

- **Deployment of system** − Once the functional and non-functional testing is done; the product is deployed in the customer environment or released into the market.

- **Maintenance** − There are some issues which come up in the client environment. To fix those issues, patches are released. Also to enhance the product some better versions are released. Maintenance is done to deliver these changes in the customer environment.

Waterfall Model – Application

Every software developed is different and requires a suitable SDLC approach to be followed based on the internal and external factors. Some situations where the use of Waterfall model is most appropriate are –

- Requirements are very well documented, clear and fixed.

- Product definition is stable.
- Technology is understood and is not dynamic.
- There are no ambiguous requirements.
- Ample resources with required expertise are available to support the product.
- The project is short..

Waterfall Model – Advantage:

The advantages of waterfall development are that it allows for departmentalization and control. A schedule can be set with deadlines for each stage of development and a product can proceed through the development process model phases one by one.

Development moves from concept, through design, implementation, testing, installation, troubleshooting, and ends up at operation and maintenance. Each phase of development proceeds in strict order.

Some of the major advantages of the Waterfall Model are as follows –

- Simple and easy to understand and use
- Easy to manage due to the rigidity of the model. Each phase has specific deliverables and a review process.
- Phases are processed and completed one at a time.
- Works well for smaller projects where requirements are very well understood.
- Clearly defined stages.
- Well understood milestones.
- Easy to arrange tasks.
- Process and results are well documented

Waterfall Model – Disadvantages:

The disadvantage of waterfall development is that it does not allow much reflection or revision. Once an application is in the testing stage, it is very difficult to go back and change something that was not well-documented or thought upon in the concept stage.

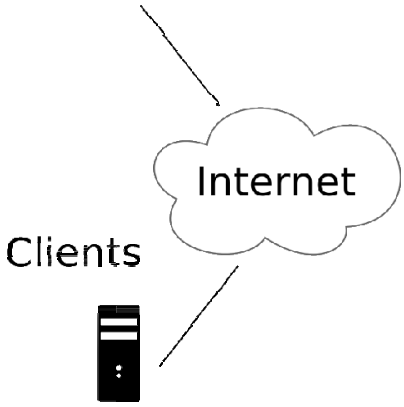
The major disadvantages of the Waterfall Model are as follows –

- No working software is produced until late during the life cycle.
- Not suitable for the projects where requirements are at a moderate to high risk of changing. So, risk and uncertainty is high with this process model.
- Cannot accommodate changing requirements.

2.3 Architecture:

Client/server architecture:

- Client/server architecture is a producer/consumer computing architecture where the server acts as the producer and the client as a consumer. The server houses and provides high -end, computing-intensive services to the client on demand. These services can include application access, storage, file sharing, printer access and/or direct access to the server's raw computing power.
- Client/server architecture works when the client computer sends a resource or process request to the server over the network connection, which is then processed and delivered to the client. A server computer can manage several clients simultaneously, whereas one client can be connected to several servers at a time, each providing a different set of Services.
- In its simplest form, the internet is also based on client/server architecture where web servers serve many simultaneous users with website data.


Internet
Clients